

RESEARCH ARTICLE

10.1002/2016WR018806

Key Points:

- Breakthrough times and effective permeabilities in DFN simulations are influenced by the adopted relationship between size and aperture
- Correlated relationship resulted in higher effective permeability values and earlier breakthrough times than networks without correlation
- Fracture geometry controls where transport occurs in a DFN while the adopted relationship controls the speed of flow and transport times

Correspondence to:

J. D. Hyman,
jhyman@lanl.gov

Citation:

Hyman, J. D., G. Aldrich, H. Viswanathan, N. Makedonska, and S. Karra (2016), Fracture size and transmissivity correlations: Implications for transport simulations in sparse three-dimensional discrete fracture networks following a truncated power law distribution of fracture size, *Water Resour. Res.*, 52, 6472–6489, doi:10.1002/2016WR018806.

Received 17 FEB 2016

Accepted 31 JUL 2016

Accepted article online 6 AUG 2016

Published online 25 AUG 2016

Fracture size and transmissivity correlations: Implications for transport simulations in sparse three-dimensional discrete fracture networks following a truncated power law distribution of fracture size

J. D. Hyman^{1,2}, G. Aldrich^{3,4}, H. Viswanathan¹, N. Makedonska¹, and S. Karra¹
¹Computational Earth Science (EES-16), Earth and Environmental Sciences Division, Los Alamos National Laboratory, Los Alamos, New Mexico, USA, ²Center for Nonlinear Studies, Theoretical Division, Los Alamos National Laboratory, Los Alamos, New Mexico, USA, ³Department of Computer Science University of California, Institute for Data Analysis and Visualization, Davis, California, USA, ⁴Applied Computer Science (CCS-7), Computer, Computational, and Statistical Sciences Division Los Alamos National Laboratory, Los Alamos, New Mexico, USA

Abstract We characterize how different fracture size-transmissivity relationships influence flow and transport simulations through sparse three-dimensional discrete fracture networks. Although it is generally accepted that there is a positive correlation between a fracture's size and its transmissivity/aperture, the functional form of that relationship remains a matter of debate. Relationships that assume perfect correlation, semicorrelation, and noncorrelation between the two have been proposed. To study the impact that adopting one of these relationships has on transport properties, we generate multiple sparse fracture networks composed of circular fractures whose radii follow a truncated power law distribution. The distribution of transmissivities are selected so that the mean transmissivity of the fracture networks are the same and the distributions of aperture and transmissivity in models that include a stochastic term are also the same. We observe that adopting a correlation between a fracture size and its transmissivity leads to earlier breakthrough times and higher effective permeability when compared to networks where no correlation is used. While fracture network geometry plays the principal role in determining where transport occurs within the network, the relationship between size and transmissivity controls the flow speed. These observations indicate DFN modelers should be aware that breakthrough times and effective permeabilities can be strongly influenced by such a relationship in addition to fracture and network statistics.

1. Introduction

Interconnected networks of fractures are the primary pathway for fluid flow and the associated transport of dissolved contaminants through low permeability rocks. These fracture networks are common in hydrocarbon extraction, aquifer storage and management, geothermal energy extraction, environmental restoration of contaminated fractured rock sites, and the disposal of spent nuclear fuel [Adler *et al.*, 2012; Berkowitz, 2002; Faybishenko, 2005; Karra *et al.*, 2015; National Research Council, 1996; Neuman, 2005]. However, the limited amount of available information leads to significant uncertainty when trying to determine the influence of fracture properties on flow and transport through the subsurface [Bonnet *et al.*, 2001]. Models used to simulate flow and transport through fractured rock address these uncertainties in different ways and each has its own advantages and disadvantages [Molz *et al.*, 2004; National Research Council, 1996; Neuman, 2005]. The two main methodologies in use are continuum models, where the rock mass is represented as a porous medium and the hydraulic conductivity is a scale-dependent correlated stochastic field, and discrete fracture network (DFN) models, where the geometry and properties of individual fractures are explicitly represented. Other models that consider the combined effect of fracture and matrix have also been recently developed [Ahmed *et al.*, 2015a, 2015b; Roubinet *et al.*, 2010; Willmann *et al.*, 2013]. Although DFN models can typically represent a wider range of transport phenomena than continuum models [Painter and Cvetkovic, 2005; Painter *et al.*, 2002], the inclusion of detailed features introduces additional layers of uncertainty because more parameters have to be calibrated [Neuman, 2005]. For example, it is possible to include in-fracture aperture variability into high fidelity DFN simulations [de Dreuzy *et al.*, 2012; Makedonska *et al.*,

2016] but constraining the in-fracture variability requires detailed knowledge of the particular rock formation.

One source of uncertainty in DFN modeling is the relationship between a fracture's size and its transmissivity. A correlation between size and aperture implicitly define a correlation between a fracture size and its transmissivity if one assumes that the aperture controls flow rate through a fracture, e.g., if the cubic law [Witherspoon *et al.*, 1980] or variant thereof is used. However, in comparison to studies on fracture length, there are relatively few on aperture distributions. Although it is generally accepted that there is a positive correlation between fracture length and aperture [Bonnet *et al.*, 2001], a functional form of that relationship remains a matter of debate. Proposed relationships between a fracture's length and its aperture include: Lévy stable [Belfield, 1998], lognormal [Charlaix *et al.*, 1987; de Dreuzy *et al.*, 2001; Margolin *et al.*, 1998], and power law [Gudmundsson *et al.*, 2001; Hatton *et al.*, 1994; Patriarche *et al.*, 2007; Vermilye and Scholz, 1995; Walmann *et al.*, 1996]. Each of these models exhibit a range of parameter values based on scale and site data. In particular, an extensive range of exponent values in power law relationships, the most widely used relationship, has been reported in the literature, cf. section 6.4.2 in Bonnet *et al.* [2001]. Moreover, any calibrated exponent is likely not universal at all scales [Hatton *et al.*, 1994; Renshaw and Park, 1997].

A direct link between fracture size and transmissivity is also difficult to obtain from field observations and proper characterization of this relationship is essential for upscaled flow and transport behavior if a correlation is adopted [Frampton and Cvetkovic, 2010]. The influence of adopting a correlation between fracture size and aperture (transmissivity) on effective network permeability was investigated using two-dimensional fracture networks by de Dreuzy *et al.* [2004] and Baghbanan and Jing [2007], but they did not consider transport processes. de Dreuzy *et al.* [2004] observed that the effective permeability of the networks was enhanced when a correlation between fracture size and aperture width was included and Baghbanan and Jing [2007] observed that the overall permeability of the DFN was controlled by large fractures that were assigned higher apertures. Joyce *et al.* [2014] and Follin *et al.* [2014] performed a site-specific study using three-dimensional networks and calibrated three different size-transmissivity models by matching specific capacities. They found that uncorrelated models had higher mean transmissivity values when compared to correlated models.

We characterize the influence that selecting a particular relationship between fracture size and transmissivity (aperture) has on transport processes in large (kilometer-scale) semigenetic sparse discrete fracture networks. We use the dfnWorks [Hyman *et al.*, 2015a] computational suite to create three-dimensional fracture networks loosely based on the fractured granite at the Forsmark site in Sweden, a potential host for spent nuclear fuel [Svensk Kärnbränslehantering AB, 2011], and resolve flow and transport therein. We select four different relationships and quantify the impact that each of them has on flow and transport observables. The four relationships we consider are: 1) a positively correlated power law, 2) a log-normal distribution around a positively correlated power law, 3) a log-normal distribution around a prescribed mean, and 4) constant (all fractures have the same transmissivity). Each of these models represents various degrees of uncertainty about the relationship between fracture size and transmissivity. Adoption of a positively correlated power law model assumes a fairly low degree of uncertainty, log-normal distribution around a positively correlated power law includes a stochastic term to account for variability of transmissivities in fractures of the same size, and the log-normal distribution around a prescribed mean assumes a high degree of uncertainty asserting that there is no correlation between the fracture size and transmissivity; the constant relationship is used for comparison. Their influence on observables is compared in terms of the effective permeability of the network, transport properties of a conservative solute passing through DFN, and where transport occurs in the network.

We observe that networks with a correlated relationship have consistently higher effective permeability values and earlier breakthrough times than networks without correlation. Specifically, fracture network geometry plays a principal role in determining where transport occurs within the network and the adopted relationship between fracture size and transmissivity controls the speed of flow and transport times. The results suggest the breakthrough times and effective permeabilities are influenced by the adopted relationship in addition to fracture and network statistics.

2. Methods

In the DFN approach, geologic field investigations are used to create a network of fractures where the geometry and properties of individual fractures are explicitly represented as $N - 1$ dimensional objects in

an N dimensional space, e.g., lines in two dimensions or planar polygons in three dimensions. Fractures in the network are assigned a shape, location, and orientation based on distributions obtained in a site characterization. Once a network is constructed, the individual fractures are meshed for computation and the flow equations are numerically integrated on the resulting computational mesh. In most DFN methodologies, the matrix is considered impervious due to its low permeability [Trimmer *et al.*, 1980]. Examples of the various DFN methodologies and their applications are found in Cacas *et al.* [1990]; de Dreuzy *et al.* [2004, 2012]; Dershowitz [2014]; Erhel *et al.* [2009]; Hyman *et al.* [2015b]; Ji *et al.* [2011]; Mustapha and Mustapha [2007]; Pichot *et al.* [2010, 2012]; and Xu *et al.* [2006].

2.1. dfnWorks

We use the computational suite `DFNWORKS` [Hyman *et al.*, 2015a] to generate each DFN, solve the steady-state flow equations and determine transport properties therein. `DFNWORKS` combines the feature rejection algorithm for meshing (FRAM) [Hyman *et al.*, 2014], the LaGriT meshing toolbox [LaGriT, 2013], the parallelized subsurface flow and reactive transport code PFLOTRAN [Lichtner *et al.*, 2015], and an extension of the WALKABOUT particle tracking method [Makedonska *et al.*, 2015; Painter *et al.*, 2012]. FRAM is used to generate three-dimensional fracture networks. LaGriT is used to create a computational mesh representation of the DFN in parallel. pflotran is used to numerically integrate the governing flow equations. WALKABOUT is used to determine pathlines through the DFN and simulate solute transport. Details of the suite, its abilities, applications, and references for detailed implementation are provided in Hyman *et al.* [2015a].

The choice to use `DFNWORKS` for the problem at hand is born from the ability of `dfnWorks` to resolve complicated flow near fracture intersections and carry out direct particle tracking with computational efficiency. Variability of transmissivity between fractures can result in poorly conditioned solution matrices and high pressure gradients near the end points of fracture intersections where points of stagnation occur in the flow field. The refined isotropic conforming Delaunay triangulation near fracture intersections allows for smooth and accurate representation of the high gradients in these regions. `dfnWorks` also allows for Lagrangian based transport simulations. We rely on Lagrangian based transport observations rather than an Eulerian framework, such as an advection-dispersion equation, because of the amount of information we obtain using individual particles and the absence of numerical dispersion that is inherent in the discretization of advection-dispersion equations. Using these pathlines allows us to use the techniques of Aldrich *et al.* [2016] and identify flow channeling through the network to link fracture network properties to flow properties. Finally, the mesh representation of the fracture networks considered here contain around $\mathcal{O}(10^6)$ control volumes. PFLOTRAN uses the PETSc [Balay *et al.*, 2016] toolbox to numerically integrate the governing equations in parallel.

2.2. DFN Model

We design a semigeneric synthetic DFN based loosely on the sparsely fractured granite at the Forsmark site, Sweden, which is a potential host formation for a subsurface repository for spent nuclear fuel. The Swedish Nuclear Fuel and Waste Management Company (SKB) has undertaken a detailed investigation of the Forsmark site that identified multiple fracture sets and multiple spatial domains [Svensk Kärnbränslehantering AB, 2010]. We create a simplified semigeneric model based on the site characterization. The selected fracture sets represent rock mass between larger deterministic fracture zones, i.e., background/stochastic fractures. Our fracture networks consist of three sets of circular fractures whose orientations follow a Fisher distribution,

$$f(\mathbf{x}; \boldsymbol{\mu}, \kappa) = \frac{\kappa \exp(\kappa \boldsymbol{\mu}^T \mathbf{x})}{4\pi \sinh(\kappa)}. \quad (1)$$

Here, $\boldsymbol{\mu}$ is the mean direction vector, which can be expressed in terms of spherical coordinates, θ' and ϕ' , and $\kappa \geq 0$ is the concentration parameter that determines the degree of clustering around the mean direction. Values of κ approaching zero represent a uniform distribution on the sphere while larger values generate small average deviations from the mean direction. The Fisher distribution is sampled using the algorithm provided by Wood [1994]. Each DFN is generated in a cubic domain with sides of length one kilometer. Fracture radii r are sampled from a truncated power law distribution with exponent γ and upper and lower cutoffs (r_u, r_o),

$$r=r_0\left[1-u+u\left(\frac{r_0}{r_u}\right)^\gamma\right]^{-1/\gamma}, \quad (2)$$

where u is a random number sampled from the continuous uniform distribution on the closed interval $[0,1]$. We select a lower cutoff of 0.015 the length one side of the domain ($r_0=15$ m) and an upper cutoff over the length of the domain ($r_u=560$ m). The upper limit of fracture size reflects a technical distinction between background and deterministic fractures identified by the site characterization. Because these simulations are meant to be generic, the inclusion of deterministic features (fractures larger than 560 m) is not necessary. We have confirmed that similar results are obtained when using a lower cutoff 10 meters and 20 meters to what is presented here. We selected the required number of fractures so that percolation almost surely occurs, but the network is still sparse. In comparison to the critical percolation p_c value, defined by Bour and Davy [1997], the connected network density p , defined in de Dreuzy et al. [2012], is barely above the percolation threshold; $p=a \cdot p_c$ with $a \in [1.0, 1.1]$ for the networks considered. The average P_{32} (surface area of the fractures over the total volume) value for the fracture network that connects the inflow and outflow boundaries of this DFN is 0.057 m^{-1} . Values of the parameters used to generate the fracture networks are provided in Table 1.

2.3. Transmissivity Models

Under the assumption of aperture uniformity, flow through the fractures is equivalent to flow through two parallel plates, and the volumetric flow rate Q per unit fracture width normal to the direction of flow is given by the Boussinesq equation [Boussinesq, 1868]

$$Q=-b^3 \frac{\rho g}{12\nu} \nabla h, \quad (3)$$

where b is the fracture aperture, ρ is the fluid density (which we assume is constant), ν is the fluid viscosity, g is acceleration due to gravity, and h is the hydraulic head. This relationship between aperture and flow rate can be used to derive a similar relationship between aperture and transmissivity

$$T=b^3 \frac{\rho g}{12\nu}, \quad (4)$$

referred to as the cubic law [Witherspoon et al., 1980]. For heterogeneous fracture apertures with significant fracture roughness and/or fracture closing, measured values of transmissivity can deviate from predictions obtained using the cubic law [Konzuk and Kueper, 2004; Zimmerman and Bodvarsson, 1996]. A host of corrections and reformulations of the cubic law have been proposed. There are reformulations based on other powers of aperture such as the quadratic law $T \propto b^2$ [Uchida, 1994] and the quintic law $T \propto b^5$ [Klimczak et al., 2010]. Cvetkovic et al. [2004] and Cvetkovic and Frampton [2012] investigated the influence that adoption of either the cubic law or the quadratic law had on transport properties in three-dimensional DFN. They determined the selection of how aperture relates to transmissivity can influence transport and retention properties. Others have proposed correction factors based on geometric and empirical considerations [Renshaw, 1995; Witherspoon et al., 1980]. We only consider the cubic law because it is already established the choice of aperture-transmissivity relationship can influence transport properties. Additionally, using a similar computational setup, Makedonska et al. [2016] found that in-fracture aperture variability has little effect on transport properties in sparse three-dimensional fracture networks. In light of their findings, each fracture is assigned a uniform aperture representative of the mean value of variable apertures within a fracture plane.

Table 1. DFN Generation Parameters^a

Set	Fisher Distribution			Power Law Distribution			Fracture Density # of Fractures
	θ'	ϕ'	κ	γ	r_u/L	r_0/L	
1. (NS)	90.0°	0.0°	21.7	2.5	0.56	0.015	2093
2. (NE)	135°	0.0°	21.5	2.7	0.56	0.015	2000
3. (HZ)	360.0°	90.0°	8.2	2.38	0.56	0.015	7711

^aThe values are loosely based on fractured granite at the Forsmark repository site in Sweden [Svensk Kärnbränslehantering AB, 2010]. The domain is a cube with sides of length (L) 1000 (m). Parameters of the Fisher distribution for fracture orientation, equation (1): Mean Trend (θ'), Mean Plunge (ϕ'), and the concentration parameter (κ). Parameters for the power law distribution of fracture radii, equation (2): power law exponent (γ), upper cutoff (r_u/L) [-], and lower cutoff (r_0/L) [-]. The required number of fractures from each increased to reach the percolation threshold. Mean P_{32} (surface area of the fractures over the total volume) value for the fracture network that connects the inflow and outflow boundaries of this DFN is 0.057 m^{-1} .

[Renshaw, 1995; Witherspoon et al., 1980]. We only consider the cubic law because it is already established the choice of aperture-transmissivity relationship can influence transport properties. Additionally, using a similar computational setup, Makedonska et al. [2016] found that in-fracture aperture variability has little effect on transport properties in sparse three-dimensional fracture networks. In light of their findings, each fracture is assigned a uniform aperture representative of the mean value of variable apertures within a fracture plane.

We consider four different size-transmissivity relationships and quantify the impact that their adoption has on flow and transport observables. The four relationships are: 1) a positively correlated power law, 2) a log-normal distribution around a positively correlated power law, 3) a log-normal distribution around a prescribed mean, and 4) constant. The functional forms of the relationships between transmissivity T and radius r are the following. The first model we consider is a positively correlated power law relationship with parameters α and β ,

$$\log(T) = \log(\alpha \cdot r^\beta). \quad (5)$$

We refer to this model as *correlated*.

The second model includes a stochastic term into (5),

$$\log(T) = \log(\alpha \cdot r^\beta) + \sigma_T \mathcal{N}(0, 1), \quad (6)$$

to account for uncertainty and variability between fractures of the same size. The strength of the stochastic term is determined by the variance of a log-normal distribution σ_T and the stochastic term is an independent identically distributed random variable sampled from a normal distribution with mean 0 and variance 1, $\mathcal{N}(0, 1)$. This model results in a log-normal distribution of fracture transmissivities around a positively correlated power law mean. We refer to this model as *semicorrelated*.

The third model assumes that there is no correlation between the fracture size and transmissivity and all values are independent identically distributed random variables from a log-normal distribution with specified mean μ_T and variance σ_T ,

$$\log(T) = \mu_T + \sigma_T \mathcal{N}(0, 1). \quad (7)$$

We refer to this model as *uncorrelated*.

The fourth model represents an assumption that in addition to no relationship between r and T , and there is no variation between fractures,

$$\log(T) = \mu_T. \quad (8)$$

We refer to this model as *constant*. Equation (8) provides a control case for comparison between the other models.

Each of these models represents various degrees of uncertainty about the relationship between fracture size and transmissivity. Adoption of the perfectly correlated model (5) implicitly assumes a fairly low degree of uncertainty about the relationship between fracture size and transmissivity. While there are indications that such a relationship between size and transmissivity is useful [Dershowitz *et al.*, 2003], the deterministic formulation is an idealization where all fractures of the same size are assigned the same transmissivity. Such a model disregards all mechanical, chemical, and hydrological processes that can result in variation between transmissivity of fractures of the same size. However, the formulation is convenient because each realization of network geometry requires a single realization of the transmissivity field. Frampton and Cvetkovic [2010] calibrated the parameters of a perfectly correlated model in a DFN representation of fractured crystalline rock at Laxemar, Sweden using soft conditioning based on Posiva flow logs (PFL). The conditioning is soft in the sense that the objective is to match the distribution of borehole flows with the distribution PFL flows rather than specific data. Other research teams have also used the perfectly correlated model [Bogdanov *et al.*, 2007; de Dreuzy *et al.*, 2002; Joyce *et al.*, 2014; Wellman *et al.*, 2009]. The semicorrelated model (6) attempts to address some of the issues associated with its deterministic counterpart. While still not resolving the physical processes that cause variations between the transmissivities of fractures of the same size, it includes a stochastic term to account for these variations [Hartley *et al.*, 2006]. While this term might increase realism of the network, it is more cumbersome than the perfectly correlated model, both in terms of computational demands (multiple transmissivity field realizations for the same network geometry are required) and calibration data (the semicorrelated model requires additional support from field data to constrain its additional parameters [Frampton and Cvetkovic, 2010]). The semicorrelated model is the least common of the models considered here, even though some researchers consider it the most realistic [Follin *et al.*, 2014]. In the uncorrelated model (7), a lognormal distribution around a prescribed mean transmissivity is used to include variability between fractures. Here the mechanical, chemical, and

Table 2. Four Different Relationships Between Fracture Radius r and Transmissivity T^a

Model	Relationship	Parameters
Correlated	$\log(T) = \log(\alpha \cdot r^\beta)$	$(\alpha, \beta) = (1.3 \cdot 10^{-9}, 0.5)$
Semicorrelated	$\log(T) = \log(\alpha \cdot r^\beta) + \sigma_T \mathcal{N}(0, 1)$	$(\alpha, \beta, \sigma_T) = (1.3 \cdot 10^{-9}, 0.5, 0.7)$
Uncorrelated	$\log(T) = \mu_T + \sigma_T \mathcal{N}(0, 1)$	$(\mu_T, \sigma_T) = (-18.79, 0.8)$
Constant	$\log(T) = \mu_T$	$\mu_T = -18.79$

^aParameters are selected so that all models have the same mean and the two stochastic relationships have similar distributions.

hydrological processes are assumed principally responsible for variations between fracture transmissivities rather than assuming a correlation between a fracture's size and its transmissivity. In other words, this model is the opposite extreme of the

perfectly correlated model and the semicorrelated is a combination of the two. Examples of studies that use the uncorrelated DFN models are *Cacas et al.* [1990]; *Cvetkovic et al.* [2004]; *de Dreuzy et al.* [2001]; *Dverstorp and Andersson* [1989]; and *Kang et al.* [2015].

We begin by comparing all four models on a single DFN realization to explore what differences occur due to assumptions of correlation and the inclusion of stochastic terms. Because there is no randomness in the correlated and constant models, only one transmissivity field is generated for each network for these models. To sample the stochastic term in the semicorrelated and uncorrelated models, thirty independent transmissivity fields are generated on a fracture network geometry for each relationship. The average statistics are studied for each set, as well as those obtained for individual realizations. This results in sixty-two flow solutions for this particular fracture network. Next, we compare the correlated and constant models using thirty independent network realizations to determine what variation exists between realizations of the fracture networks. These two relationships, correlated and constant, are selected because statistics obtained from the semicorrelated and uncorrelated center around their deterministic counterparts. In sum 112 different flow fields on 31 different network geometries are considered.

The adopted parameters of the four different relationships for the single DFN realization are provided in Table 2. The parameters are selected so that the mean transmissivity of the fracture networks are all the same and that the distributions of aperture and transmissivity in models that include a stochastic term are also the same. In all cases the P_{32} values are nearly identical; differences between samples are less than 10^{-5} . The parameter values and variances are taken from those provided in Table 6-75 in the SKB report TR-10-52 [*Svensk Kärnbränslehantering AB*, 2010]. Figure 1 shows the distributions of fracture apertures and transmissivities for the correlated (blue), semicorrelated (red), and uncorrelated models (green). The values

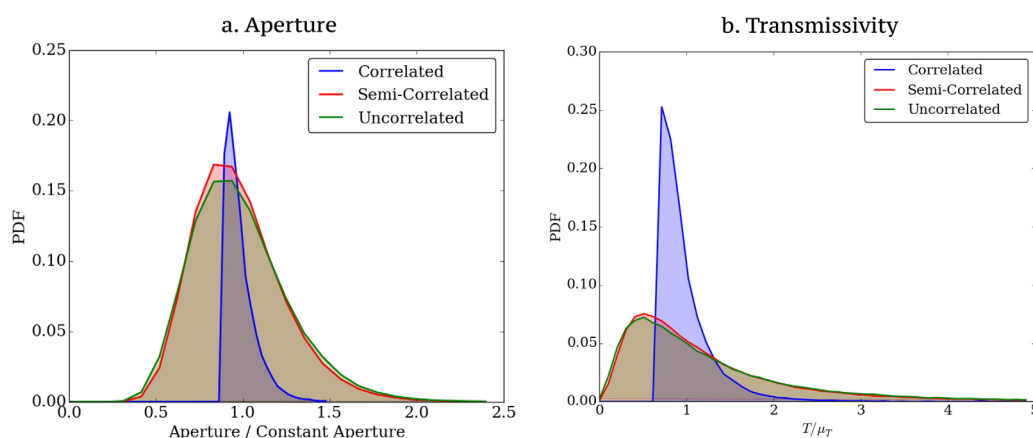


Figure 1. Empirical probability density functions of the (a) fracture apertures and (b) transmissivities for the correlated (blue), semicorrelated (red), and uncorrelated models (green). Model parameters, provided in Table 2, are selected so that the mean transmissivity of the fracture networks are all the same and that the distributions of aperture and transmissivity in models that include a stochastic term are similar. Values are normalized by the value of the constant model for comparison. (a) Even though the mean values of aperture are all the same, there is a clear difference between the distributions of the deterministic model (correlated) and the two stochastic models (semicorrelated and uncorrelated) due to the stochastic term. The stochastic term in the semicorrelated and uncorrelated models results in wider distributions with heavier tails that do not exhibit a sharp cutoff in the distribution, which occurs in the correlated case due to the truncated power law distribution of fracture radii. (b) The distributions of transmissivity for the two stochastic models differ from the correlated model, but are close to one another. The distribution of transmissivities for the stochastic models display a much fatter tail at high values of transmissivity, include more small transmissivity fractures, and have a higher variance than the deterministic model.

are normalized by the aperture of the constant model for comparison. Figure 1a shows the distribution of apertures for the correlated, semicorrelated, and uncorrelated models. Even though the mean values of aperture are all the same, there is a clear difference between the distributions of the deterministic model (correlated) and the two stochastic models (semicorrelated and uncorrelated) due to the stochastic term. In the correlated case, the apertures (and transmissivities) are determined by the power law distribution for the fracture radius, which has a hard cutoff. This lower bound results in distributions with a sharp cutoff in the distribution of apertures as well, reminiscent of a power law distribution. The stochastic term in the semicorrelated and uncorrelated models results in wider distributions with heavier tails. Figure 1b shows the distribution of transmissivities for the correlated, semicorrelated, and uncorrelated models. The distributions of the two stochastic models differ from the correlated model, but are nearly indistinguishable from one another. The distribution of transmissivities for the stochastic models have a much fatter tail at high values of transmissivity, include more small transmissivity values, and have a higher variance than the deterministic model.

2.4. Flow and Transport Simulations

To characterize the influence that adopting any of these models has on flow and transport through each network we resolve steady-state single phase flow in each network and then simulate transport of a nonreactive conservative solute in DFN using by a cloud of passive tracer particles, i.e., using a Lagrangian approach. We rely on Lagrangian based transport observations rather than an Eulerian framework, such as an advection-dispersion equation, because of the amount of information we obtain using individual particles and the absence of numerical dispersion that is inherent in the discretization of advection-dispersion equations. Using these pathlines we employ the techniques of Aldrich *et al.* [2016] to identify flow channeling through the network and link fracture network properties to flow properties. Dirichlet boundary conditions are imposed on the top (1 MPa) and bottom (2 MPa) of the domain to create a gradient aligned with the vertical axis, and no flow boundary conditions are enforced along lateral boundaries. We do not include gravity in these simulations. The distribution of pressure in the fracture network is solved throughout the domain assuming a single-phase flow at steady-state. Figure 2 shows an image of a single DFN realization with fractures colored by the steady-state pressure solution. The imposed pressure gradient is aligned with

the vertical axis, z , and thus the primary direction of flow is also in the vertical direction.

In each steady-state flow field, particles are inserted uniformly across all fractures at the inlet plane ($z = 0$) of each network to simulate transport. Each particle has a unique initial position $\mathbf{x}_i = (x_i, y_i, 0)$ for $i = 1, \dots, N$, where N is the total number of particles, on the fractures that intersect the inlet plane. The trajectory of the i th particle starting at $\mathbf{x}_i$ is defined by the pathline $\mathbf{p}_i$, which is the solution to initial value problem

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}(\mathbf{x}(t)), \quad \mathbf{x}(0) = \mathbf{x}_i. \quad (9)$$

Here, $\mathbf{v}$ is the Eulerian velocity vector with components $\mathbf{v} = [u, v, w]$ defined at every point in the fracture network. The length of the pathline ℓ is used to parameterize the spatial and

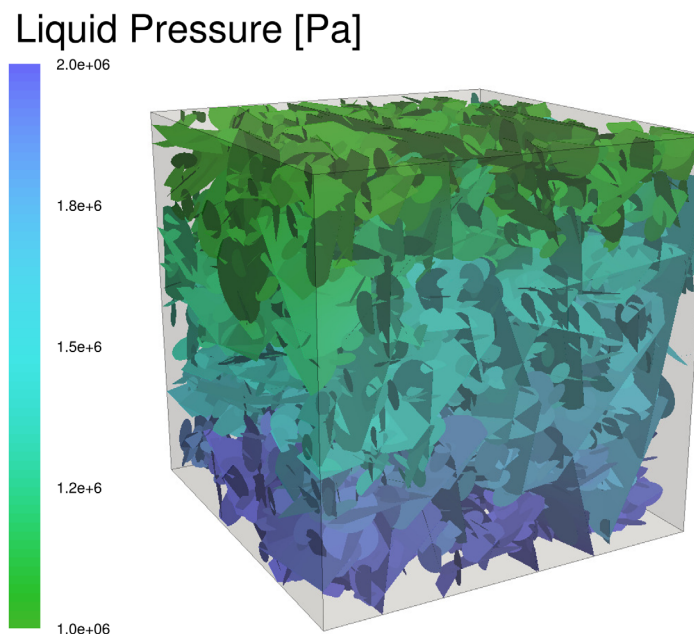


Figure 2. Steady-state pressure solution inside of a DFN. The DFN shown here is loosely based on the fractured granite in Forsmark, Sweden, a potential host for used nuclear fuel. Fracture transmissivity is typically assumed to be positively correlated with fracture radius, which follows a truncated power law distribution. Using such fracture networks we investigate the influence that relationships presented in Table 2 have on flow and transport properties.

temporal coordinates of the particle. Explicitly, $\mathbf{p}_i(\ell) = [x(\ell), y(\ell), z(\ell), t(\ell)]$ for $\ell > 0$ and $\mathbf{p}_i(0) = [x_i, y_i, 0, 0]$. At any point all along $\mathbf{p}_i(\ell)$ we can observe the aperture of a fracture $b(\ell)$. Under the assumptions of steady-state flow and in the absence of local dispersion, particles move along their respective pathlines and arrive exit the domain a distance z away from the inlet at the point $\mathbf{p}_i(\ell') = [x(\ell'), y(\ell'), z, t(\ell')]$.

Assuming the rock matrix is porous and fully saturated with water each particle transported along its trajectory through the fractures may be temporally transferred to the matrix by diffusion, sorption, or a combination of the two. Cvetkovic *et al.* [1999] derived the probability density function of the particle resident time to travel a distance z under these conditions and determined that it depending upon two Lagrangian based parameters (τ : particle breakthrough time and β : transport resistance/retention parameter) and one material property group that combines the matrix porosity, effective diffusivity, and retardation coefficient in the matrix. The Lagrangian integral along each trajectory can be used to compute the breakthrough time $\tau_i(z)$ and the retention parameter $\beta_i(z)$ for each particle,

$$\tau_i(z) = \int_0^{\ell'(z)} \frac{d\xi}{w(\xi)}, \quad \beta_i(z) = \int_0^{\ell'(z)} \frac{2d\xi}{b(\xi)w(\xi)} \quad (10)$$

to reach the exit plane. Here, $w(\ell)$ is vertical Eulerian velocity of a particle along its pathline. The transport time τ is the time it takes for a solute particle to pass through the network and the transport resistance β controls mass transfer with the matrix along a flow path. These correlated Lagrangian variables are key parameters that control advection, diffusive mass transfer and surface sorption of dissolved solutes. The importance of τ and β transport through in individual fractures and fracture networks has been underscored in numerous studies [Cheng *et al.*, 2003; Cvetkovic *et al.*, 2004; Cvetkovic and Frampton, 2012; Frampton and Cvetkovic, 2011; Painter *et al.*, 2002]. Frampton and Cvetkovic [2011] simulated flow and transport in the calibrated model constructed by Frampton and Cvetkovic [2010], discussed in section 2.3. They analyzed distributions of τ using the truncated one-sided stable distribution [Cvetkovic, 2011]. They observed significant flow channeling where a large portion of the solute transport occurred within larger fractures with high transmissivities. They adopted a perfectly correlated power law relationship between fracture size and transmissivity. Cvetkovic and Frampton [2012] used the same setup and studied β in addition to τ . One of the aspects they consider is the correlation of τ with β . It has been common to assume that the two are linearly related

$$\beta(\tau) = s_f \tau \quad (11)$$

where s_f (m^2/m^3) is the active specific surface area of the rock [Cvetkovic *et al.*, 1999; Cvetkovic and Frampton, 2012]. Active refers to the proportion of the fracture surface area where flow is occurring and solute/matrix interaction possible as opposed to passive specific surface area, e.g., P_{32} . This value is particularly important when considering reactive transport as it controls the amount of area available for a dissolved solute [Steefel and Lichtner, 1994]. We use distributions of β and τ to approximate s_f and investigate its dependence on the adopted size-transmissivity model.

Given a fixed total mass represented by the particles M , the mass associated with each particle m_i can be assigned using two different methods, flux-weighted and resident based. Each of these methods are meant to represent different physical scenarios. Based on the analysis of Frampton and Cvetkovic [2009] and Hyman *et al.* [2015b], whom studied how the choice of injection mode influences the distribution of travel times in three-dimensional fracture networks, we opt to use flux-weighted injection because it provides more stable statistics in terms of distributions of travel times and retention.

The mass represented by each particle and the breakthrough time at each control plane can be combined to compute the total solute mass flux $F(t, z)$ that has crossed the outlet plane a distance z away from the inlet at a time t ,

$$F(t, z) = \frac{1}{M} \sum_i m_i H[t - \tau_i(z)] \quad (12)$$

In (12), $H(t)$ is the Heavyside function where $H(t') = 1$ for $t' > 0$, and $H(t') = 0$ otherwise. Equation (12) is the integral form of the cumulative distribution function (CDF) of solute passing through a control plane. We also consider the expression $1 - F(t, z)$, the complementary cumulative distribution function (CCDF), which

when plotted on a log-log scale can be used to determine power law scaling of the tail of the $F(t, z)$. A similar equation is used to compute the CDF and CCDF for β by replacing τ_i with β_i .

In addition to analyzing distributions of breakthrough and retention observations, we identify backbones in the fracture network using the methods of Aldrich *et al.* [2016]. For flow and transport simulations in a DFN, we define a backbone as a connected subset of fractures that forms a path in the network on which a substantial portion of flow and transport occurs. This definition of a backbone is different than the classical definition used in percolation theory. To identify subsets of fractures that compose a backbone, we generate a flow topology graph (FTG) for analysis, which is dual to the fracture network topology. In this graph, each node represents a fracture in the DFN, and an edge connecting nodes represents an intersection between two fractures. Attributes of the transport simulation, which are defined using particle trajectories, are embedded into the corresponding nodes and edges of the FTG to construct a directed graph. Using a graph representation allows us to apply an optimal path algorithm that identifies possible backbones in the network. The topology of the graph and the fracture network are identical, thus paths extracted in the FTG are represented by connected subsets of fractures in the DFN. The analysis algorithm balances three metrics in order to produce an optimal path that we define as a *candidate backbone*. The first metric is maximizing the number of particles that travel along a path, as backbones should contain a significant portion of the local flow. The second metric minimizes the average tortuosity of particles along a path, as backbones should create a relatively direct path to the network outlet. The final metric minimizes the number of fractures in a path, which ensures that the path is relatively simple. We apply the algorithm both globally and on local subsets to detect multiple backbones in the system. Once these paths are extracted, we confirm the validity of each candidate by comparing them to clustered particle trajectories, which represent the trace of particles advected through the system. The trajectories are clustered using a topological algorithm that groups particles by the similarity of their paths in terms of the ordered set of fractures each particle traverses. Large numbers of particles in the same group represent channelization in the network. By comparing the extracted paths to these clustered particles, we evaluate the candidates and identify paths, which represent backbones for each simulation. If we identify multiple backbones we rank them in a hierarchy according to the three previous metrics. For details and a full discussion of the algorithms briefly described here see Aldrich *et al.* [2016].

3. Results

In this section, we report how the adopted relationships between fracture size and transmissivity influence flow and transport behavior in terms of the effective permeability of the network, particle breakthrough times and transport resistance curves, and network backbone. We begin with a single DFN realization to focus our discussion and presentation of results and then report observations for multiple fracture networks.

3.1. Effective Permeability

To estimate the effective permeability of each network in the direction of flow, we compute the Darcy velocity at the outlet plane and then invert Darcy's law using the imposed pressure gradient and domain size. Values of mean, variance, minimum and maximum for the samples are reported in Table 3. For comparison, effective permeability values are normalized by the effective permeability of the constant fracture network. Figure 3 shows a scatter plot of the semicorrelated and uncorrelated values along with a black dotted line of the effective permeability of the constant fracture network and a blue dotted line for the effective permeability of the perfectly correlated network.

Table 3. Effective Permeabilities^a

Model	Mean	Variance	Minimum	Maximum
Correlated	2.56			
Semicorrelated	2.62	0.13	3.35	2.03
Uncorrelated	0.98	0.04	1.46	0.66

^aValues are normalized by the effective permeability of the constant fracture network for comparison.

The effective permeabilities of the networks are within a factor of two to three of one another. The perfectly correlated model resulted in a value of effective permeability two and one-half times that of the constant network even though the mean values of transmissivity in the two networks are the same. The reported value of the semicorrelated model center around that of the

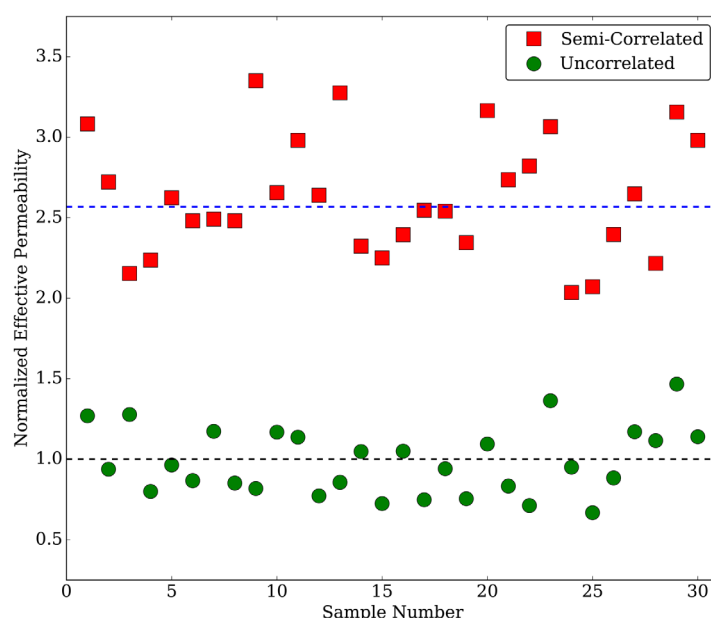


Figure 3. Effective permeability of the fracture networks. Values are normalized by the effective permeability of the constant fracture network for comparison (black dotted line). The effective permeability for the perfectly correlated model (blue dotted line) is 2.56 times that of the constant fracture network. Values for the uncorrelated model center around the value of the constant model. Values for the semicorrelated model are higher than the constant and center around the value of the correlated model and exhibit more variation than the uncorrelated model.

perfectly correlated value, which is expected because the semi-correlated relationship is based on the perfected correlation relationship, and the uncorrelated networks' effective permeabilities center around that of the constant network, which is expected because the uncorrelated relationship is based on the constant relationship. There is more variation between realizations using the semicorrelated relationship when compared to the uncorrelated relationship.

The observed difference between effective permeability values of the correlated/semi-correlated and constant/uncorrelated models demonstrates that correlation between fracture size and transmissivity influences the effective permeability of a DFN. Considering that the mean values of the fracture based transmissivities

are the same and the distributions of the two stochastic models are nearly identical, cf. Figure 1b, it appears that correlations between fracture size and transmissivity impact upscaled transport properties in a manner that cannot be known *a priori* when considering only the distributions of aperture and transmissivity. In other words, fracture network geometry also plays a key role in determining upscaled flow properties. *de Dreuzy et al.* [2004] observed similar results in two-dimension DFN simulations, but the impact on transport, which we consider in the next section, was not investigated.

3.2. Travel Time and Transport Resistance Distributions

Figure 4a reports the empirical cumulative distribution function (CDF) for the particle travel times (breakthrough curves) computed using equation (12). Figure 4b reports the complement of the cumulative distribution function (CCDF) on a log-log plot. Semitranslucent lines represent individual realizations and thick lines are the aggregate of all realizations for a given relationship. DFN with semicorrelated relationships are red, uncorrelated: green, correlated: blue, and constant: black. For comparison, time (ordinate) is rescaled by the 50% breakthrough of particles in the constant fracture network.

Breakthrough times for the correlated and semicorrelated networks are significantly earlier than those observed for constant and uncorrelated networks. Table 4 reports the 50% breakthrough times and exponent of a power law fit to CCDF tail. The earliest breakthroughs are observed in semicorrelated networks and the latest breakthrough times are observed in the uncorrelated model. The aggregate values of the semicorrelated networks are very close to the correlated values, similarly for the uncorrelated and constant networks. This is not surprising however, because the semicorrelated relationship is based on the correlated and the uncorrelated relationship is based on the constant model. However, at early times the stochastic models have slightly earlier breakthroughs than their deterministic counterparts. This deviation is more pronounced in the uncorrelated/constant case. There are particular realizations of the semicorrelated model where the 50% breakthrough is later than particular realizations of the uncorrelated model. While it is already known that the effect of transmissivity variability on the DFN scale can be significant, these distributions provide evidence that the effect it is increased when transmissivity is correlated with fracture size. The exponent of the power laws fit to the tail of the distributions are all approximately the same ≈ 2 . These observations suggest that a correlated relationship between fracture

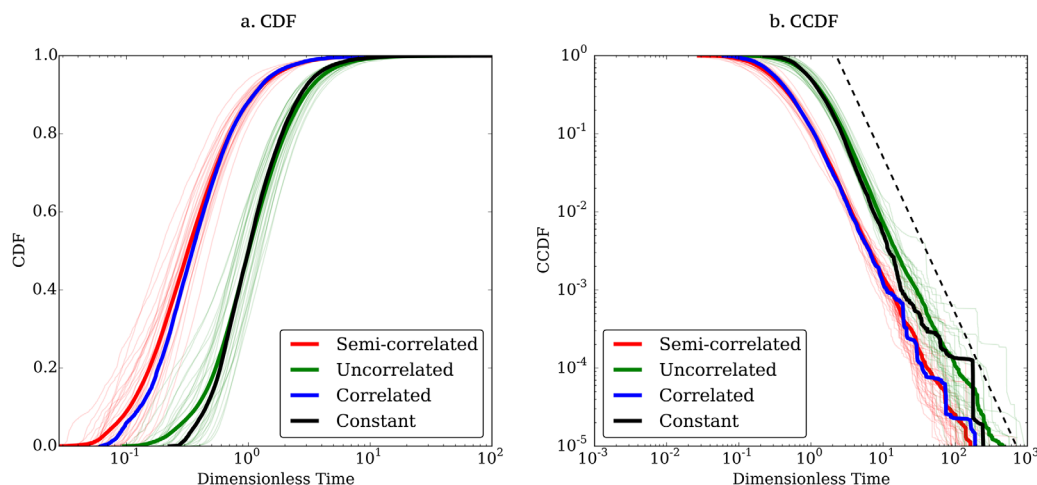


Figure 4. Breakthrough times (τ). Semicorrelated relationships between fracture size and transmissivity are red, uncorrelated: green, correlated: blue, and constant: black. For comparison, time on the ordinate is rescaled by the 50% breakthrough of the constant fracture network. (a) Cumulative distribution of travel times. Breakthrough times for the correlated and semicorrelated networks are much earlier than that observed for constant and uncorrelated. The earliest breakthroughs are observed in semicorrelated networks, and the latest breakthrough times are observed in the uncorrelated model. The aggregate values of the semicorrelated networks are very close to the correlated values, similarly for the uncorrelated and constant networks. (b) Complement of cumulative distribution of travel times. All tails exhibit the similar exponents of power law scaling (Table 4) suggesting that the adopted relationship does not influence long-term transport behavior in the DFN. The dashed black line is a power law with an exponent of two.

size and transmissivity leads to earlier breakthrough time but the scaling of the distribution at later times is not significantly influenced.

Considering each fracture network as a homogenous permeability fields with an effective permeability k_e , a straightforward calculation using Darcy's law shows that increase of effective permeability by a factor of 2.5, as reported in Table 3 for the correlated cases, results in a decrease of travel time by 60%, i.e., breakthrough in the higher permeability fields is 0.4 times the breakthrough in the lower permeability field. This analytical value is quite close to the observed values that are reported in Table 4 where the median breakthrough of particles in the perfectly correlated network is 0.34 times the breakthrough of that observed in the constant network. This similarity is not surprising because both the effective permeability and breakthrough curves are upscaled observables of the networks. However, the reported value is slightly smaller than what is expected analytically indicating that particles in the correlated networks move faster than what would be expected in an equivalent continuum representation.

In general, the CDF and CCDF of β are rather similar to those of τ and thus are not shown. There are a few differences worth commenting on however. The mean values of the semicorrelated networks is rather close to that of the correlated network, similarly for the constant and uncorrelated networks, but less so than for values of τ . Deviations of the stochastic models are more pronounced in the CDF of β than of τ . Distributions of β are more sensitive to changes in aperture than τ , because b is used in the computation of β . Therefore variations in aperture along the pathlines are more reflected in plots of β than of τ , which likely leads to these deviations. At early times, the models that include a stochastic component have slightly earlier breakthroughs than their deterministic counterparts. All tails exhibit roughly the same power law scaling

suggesting that the adopted fracture-size transmissivity relationship influences the early retention times more than long-term transport behavior in the DFN. Table 4 reports the 50% breakthrough times and exponent of a power law fit to CCDF tail, which are nearly identical to those reported for τ .

Table 4. 50% Breakthrough Times and Exponent of Power Law Fit for Particle Travel Time and Retention Parameter^a

		Correlated	Semicorrelated	Uncorrelated	Constant
τ	50% Breakthrough	0.34	0.31	1.01	1.00
	power law Exponent	1.95	1.90	2.00	2.18
β	50% Breakthrough	0.27	0.24	1.02	1.00
	power law Exponent	1.87	1.84	1.96	2.18

^aTimes are normalized by the 50% breakthrough time for the constant network.

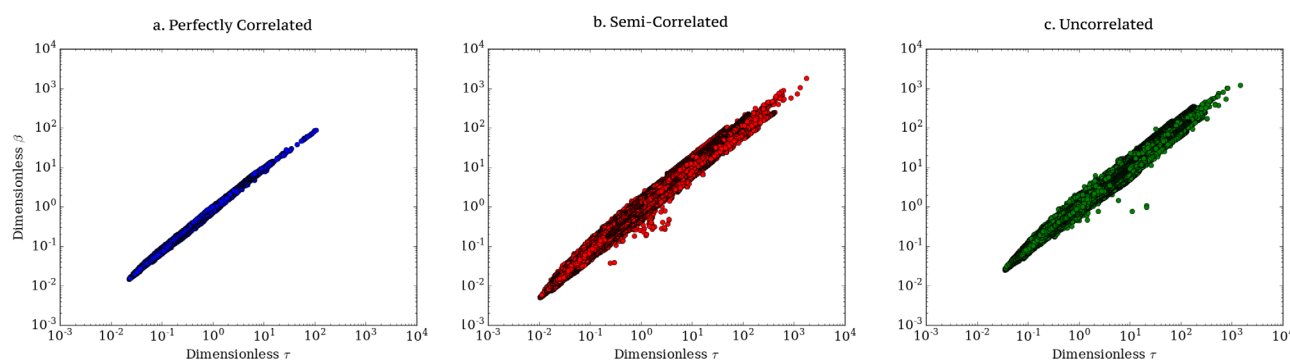


Figure 5. Scatter plots of β as a function of τ for the (a) perfectly correlated model, (b) the semicorrelated model, and (c) the uncorrelated model. Values of τ and β are normalized by the median values from the constant model for comparison. Best linear fits to the scatter plots provide an estimate of the active specific surface area associated with each model.

3.3. Active Specific Surface Area

Although the distributions of τ and β are quite similar, their relationship with one another varies with the adopted size-transmissivity model. Figure 5 shows scatter plots of β as a function of τ for the (a) perfectly correlated model, (b) the semicorrelated model, and (c) the uncorrelated model. Values of τ and β are normalized by the median value of the constant model for comparison. The constant model is not included because $\beta = 2\tau/b$, as b is constant. A linear relationship between the two is formalized by the active specific surface area defined in (11). A linear model fits the data well; R^2 greater than 0.9 for all size-transmissivity relationships. The specific surface area associated with each model is approximated by taking the median value of β/τ across all particles. Relative to the active surface area of the constant model, the active surface area of perfectly correlated model is 0.82 (variance of 0.003), 0.84 (variance of 0.02) for the semicorrelated, and 1.04 (variance of 0.02) for the uncorrelated models. In other words, the active surface area in the two correlated models is around 15–20% lower than the uncorrelated models. Such values indicate that size-transmissivity relationship could influence transport mechanisms like diffusion, retention and fracture-matrix exchanges because the active surface area available for these processes depends on the adopted model.

3.4. Network Backbone

Backbones of a fracture network are connected subsets on which a majority of flow and transport occurs. Backbones of each network are determined using the flow topology graph method described in section 2.4. Figure 6 shows the two primary backbones of the (a) perfectly correlated and (b) constant fracture networks. The primary backbones in the semicorrelated networks and uncorrelated networks are similar to those shown here (not included). The backbones are made up of relatively large fractures, which are generally aligned with the direction of flow (bottom to top). One of the backbones is colored red and the other blue. In the case of the red backbones, all the fractures are the same. They differ however, in the case of the blue backbone. For the perfectly correlated case there are more large fractures due to their higher transmissivities. In the constant network, the fractures are smaller and more aligned with the direction of flow (top to bottom). This result suggests that the fractures on which a majority of transport through the networks occurs is primarily determined by network geometry and the adopted relationship between fracture size and transmissivity is a secondary effect. At higher values in the backbone hierarchy the fractures that make up the backbones are different from one another in different realizations.

While the set of backbone paths extracted for the perfectly correlated and constant aperture simulations are similar, the amount of flow traveling along them is different. In both cases, the backbones are composed of a few very large fractures that traverse the network, several medium sized fractures that are typically oriented in the direction of flow, or a combination of the two. For the perfectly correlated fracture network, 49% of the particles pass through the top five backbones while the backbones only contribute 15% of the total surface area for the network. Particles disperse more in the fracture network with constant aperture than in the perfectly correlated case. Only 29% of the total particles travel over the top five

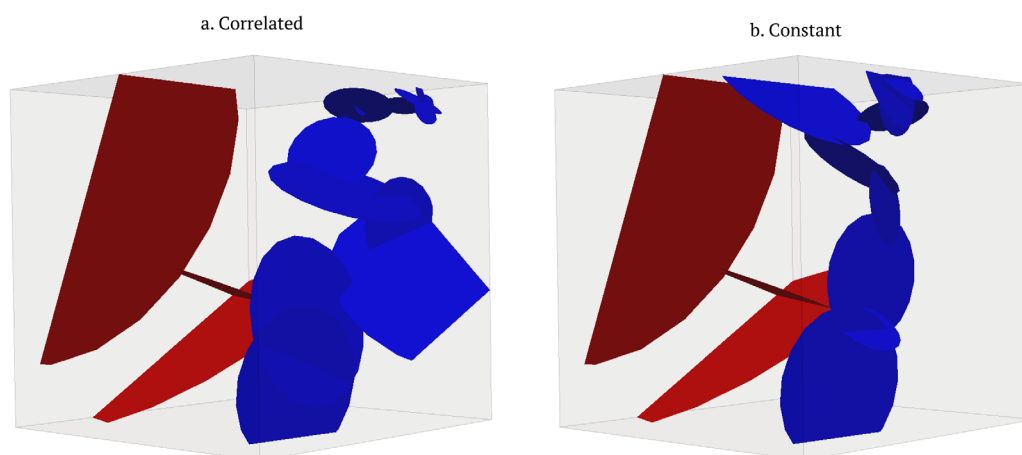


Figure 6. Primary fracture network backbones for the networks with (a) perfectly correlated and (b) constant fracture size and transmissivity relationships. Backbones in the fracture network are connected subsets of the fracture network on which the majority of transport occurs. The fractures that make up the primary backbones are nearly the same for the perfectly correlated and the constant networks, indicating that network geometry is the principal factor in flow channeling and the adopted relationship between fracture size and transmissivity is a secondary effect. Backbones are composed of larger fractures in line with the primary direction of flow.

backbones and the backbones contribute 20% of the total surface area for the network. The FTG graph analysis also indicates that increased flow channeling results from the correlation between fracture size and transmissivity. In the correlated case, only 43% of the fractures in the network are touched by a particle during transport simulations. In the semicorrelated case, the average is 41% with standard deviation of $\approx 1\%$. In contrast, the average number of fractures touched by a particle in the uncorrelated and constant networks are 50% and with standard deviation of $\approx 1\%$ and 52%, respectively. These values indicate that flow channeling occurs in all of the fracture networks and is more pronounced when correlations between fracture size and transmissivity are included.

3.5. Multiple Geometry Realizations

We generate thirty additional fracture networks using the same statistics (Table 1) and compare breakthrough curves assuming either a correlated or constant fracture size-transmissivity relationship. This results in an additional sixty flow solutions. These two relationships, correlated and constant, are selected because breakthrough curves of the semicorrelated and uncorrelated center around these values respectively, cf. Figure 4. Therefore general differences in overall flow and transport for the stochastic models in these DFN can be inferred by their deterministic counterparts.

Figure 7a shows the CDF for the ten DFN realizations and (b) shows the CCDF. One hundred thousand particles are tracked through each DFN for both the constant and correlated cases. Time is normalized by the 50% breakthrough time of the aggregate constant breakthrough times. Semitransparent lines are single realizations and thick lines are the aggregate of all realizations. Differences between realizations are more pronounced in the correlated networks. Similar to what is observed in Figure 4, these plots indicate that including correlations between size and transmissivity results in significantly earlier breakthroughs than if correlations are not included. The aggregate 50% breakthrough time of the trajectories through correlated networks is about 50% faster than through the constant networks. The standard error for the 50% breakthrough time for the constant networks is 0.018 and 0.013 for the perfectly correlated indicating that the observed differences in 50% breakthrough times are not a result of an insufficient number of realizations. The CCDF for these curves is shown in Figure 7b and demonstrates that these differences persist at later times. The dashed black line is a power law with exponent of two and shows that both relationships exhibit similar power law scaling suggesting that scaling of the distribution at later times is not significantly influenced by the adopted relationship. These plots demonstrate that the large discrepancies in travel times resulting from the adopted size and transmissivity relationships do not depend on the particular realization.

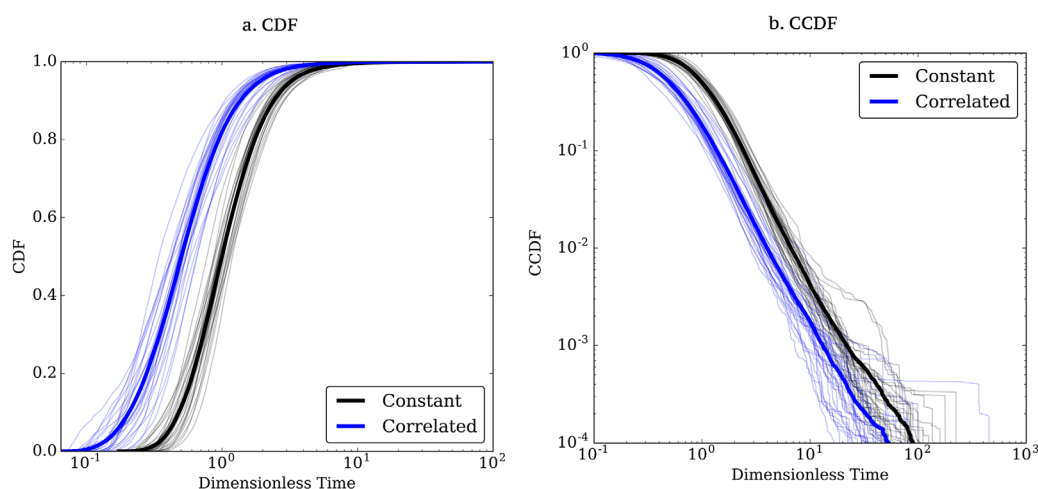


Figure 7. Breakthrough time distributions for thirty independent DFN realizations. Time is normalized by the 50% breakthrough of the aggregate constant breakthrough times. Semitransparent lines are single realizations and thick lines are the aggregate of all realizations. For the same fracture network geometry, correlating transmissivity with size results in earlier breakthrough times when compared to simulations where no correlation is adopted.

4. Discussion

We have characterized the influence that selecting a particular relationship between fracture size and aperture has on flow and transport processes in a large-scale sparsely fractured three-dimensional networks. We compared four different fracture size-transmissivity relationships to investigate the potential impact that each model has on flow and transport. Comparisons are made in terms of the network effective permeability, transport properties including travel time, retention, active surface area, and fracture network backbone structure.

The principal observation of this study is that adopting a correlation between fracture size and transmissivity in a three-dimensional DFN can result in significantly earlier breakthrough times and higher effective permeabilities than if a correlation is not included. We also found that fracture network geometry has a stronger influence in determining where transport occurs within a sparsely fractured DFN network than the adopted relationship between fracture size and transmissivity, which controls the speed of flow and transport.

The following list summarizes our key observations:

1. Even though the mean values of the fracture transmissivities are the same for all models and the distribution of transmissivities of the two stochastic models are nearly identical (Figure 1) the perfectly correlated and semicorrelated relationships resulted in consistently higher effective permeability values than the constant and uncorrelated relationships (Figure 3 and Table 3). These results suggest that correlations between fracture size and transmissivity influence upscaled transport properties in a manner that cannot be predicted by solely considering distributions of aperture and transmissivity due to the constraints put on the flow field by the fracture network geometry.
2. Breakthrough times for the correlated and semicorrelated networks are consistently and significantly earlier than those observed in the constant and uncorrelated models (Figure 4; Table 4 and Figure 7). These observations suggest that flow moves faster through the entire network when larger fractures are assigned higher values of transmissivity.
3. The distribution of particle travels times in the correlated and semicorrelated networks are very close, as are the constant and uncorrelated networks (Figure 4a and Table 4). This is not surprising however, because the semicorrelated relationship is based on the correlated and the uncorrelated relationship is based on the constant model. Models that include a stochastic component, semicorrelated and uncorrelated, have earlier breakthroughs than their deterministic counterparts.
4. The tails of the travel time and retention parameter distributions exhibit the same power law scaling behavior, with exponents that are close to 2 (Figure 4b and Table 4). This similarity suggests that the

adopted fracture-size transmissivity relationship does not have a pronounced influence on long-term transport scaling behavior.

5. The active surface area, computed using τ and β , for the two correlated models is between 15 to 20% lower than the uncorrelated models (section 3.3). These differences could have significant impact on reactive transport modeling because the active specific surface area indicates the amount of surface area available for a dissolved solute.
6. Flow channeling occurs in all of the fracture networks. If correlations between fracture size and transmissivity are included, then it is more pronounced (section 3.4).
7. Fractures that make up the primary backbones of the networks are similar regardless of the adopted relationship between fracture size and transmissivity. (Figure 6) Fracture network backbones are composed of relatively large fractures that are primarily aligned with the direction of flow. These results indicate that the fractures where the majority of transport through the network occurs are primarily determined by the network geometry and the adopted relationship between fracture size and transmissivity is a secondary effect with respect to determining where flow occurs.

5. Conclusions

A possible explanation for the deviation between the models in terms of flow and transport observables is that the majority of flow and transport occurs in big fractures due to geometrical induced channeling. In the case of the perfectly correlated and semicorrelated models this channeling is further encouraged by the reduced resistance offered by the higher transmissivity values associated with these large fractures. This flow channeling results in higher effective permeability values and faster breakthrough times for these two relationships. Another possibility is that including a correlation between fracture size and aperture influences the overall fracture volumes, which could partially account for the observed differences. In the case of the correlated relationships, the total volume of the fracture network is higher than the uncorrelated, averaging $1.5 \cdot 10^{-4} \text{ m}^3/\text{m}^3$ compared to $1.2 \cdot 10^{-4} \text{ m}^3/\text{m}^3$. Recall that the mean transmissivity and P_{32} values were constant between realizations, but not total fracture volume. This difference can thus explain some of the differences between the breakthrough curves. It is more likely that the increased volume of the large fractures, resulting in higher transmissivity in the principal flow regions, is more responsible for the differences, than the over all increase in volume fraction.

Within any DFN simulation several assumptions and conceptualizations have to be invoked. For this study, we assumed that: fracture aperture is constant within each fracture, transmissivity is controlled by the cube of the aperture, a relatively low variability of transmissivity is considered, particles are inserted according to flux weighting, the network is sparse, fractures are circles, fracture radii follow a truncated power law distribution with hard upper and lower cutoffs, and fracture centers are uniformly distributed. Some of these assumptions, such as uniform fracture aperture; network sparsity; and the adoption of the cubic law, could influence the magnitude of the observed differences due to adoption of a particular size-aperture relationship. For example, it is possible that discrepancies between size-transmissivity models might be exacerbated/reduced if different aperture-transmissivity relationships are considered. The adoption of the cubic law has been shown to influence the computation of τ and β [Cvetkovic *et al.*, 2004; Cvetkovic and Frampton, 2012] and combining our results with those of Cvetkovic and Frampton [2012], who found that adopting the cubic law can lead to early breakthrough up to 4 times faster than those obtained using the quadratic law, it could be possible that the same fracture network geometry could produce mean breakthrough times that are an order of magnitude different from one another, e.g., a perfectly correlated model using the cubic law compared to a uncorrelated model using the quadratic law. Whether these assumptions combine in a linear or nonlinear fashion is an open area of research. A better understanding of both the hydraulic relationship between aperture and transmissivity and size-transmissivity relationships and their joint sensitivity is imperative for robust modeling using discrete fracture networks. Similarly, although Makedonska *et al.* [2016] found that in-fracture variability had little effect on global transport properties in similar networks it is possible that different internal aperture structures representative of various physical processes such as channelization of apertures due to erosion, could have more significant impacts when combined with a correlation relationships between size and aperture. Furthermore, our problem setup was designed so that distributions of transmissivity were identical, albeit with low variance. While the similarity between the constant and uncorrelated networks under such setup is not terribly remarkable, it is remarkable that a single

fracture network geometry with identical transmissivity distributions (Figure 1) but different size-transmissivity correlation relationships (equations (5–8)) can produce differences of mean breakthrough times that can be up to half an order of magnitude different (Figure 4). While it is already known that the effect of transmissivity variability on the DFN scale can be significant, this study provides evidence that the effect is increased when transmissivity is correlated with fracture size.

Regardless of the assumptions made here, the observed differences between transport properties serves as a warning for DFN modelers. If fracture network statistics are such that geometrically induced flow channeling through a few large fractures parallel to the primary direction of flow are common, then the differences resulting from the use of different fracture-size transmissivity relationships will likely be more pronounced than networks where fractures sizes are relatively uniform. Thus the DFN modeler should take care to not only consider the distribution of transmissivities in the network, but also the distributions of fracture size, transmissivity, and orientation relative to the flow domain if they want different size-transmissivity relationships to result in similar flow behaviors. In the absence of good estimates of aperture (T) distributions, then using flow measurements is the only viable option to calibrate DFN parameters. Another project comparing these models using data from the Laxemar site, similar to the study of *Frampton and Cvetkovic* [2010], could help shed light on the variations induced by adoption of one these models. However, calibration of model parameters by matching specific capacities and flow has resulted in higher values of mean transmissivity for uncorrelated models when compared to correlated models [Joyce et al., 2014; Follin et al., 2014]. The calibrated values resulted in a significant difference of mean log transmissivities between models. In light of the findings reported here, the higher values likely compensate for the increased flow channeling encouraged by the correlated relationships by globally increasing the transmissivity of the entire model to match flow parameters. While this modification might decrease the difference between model outputs in terms of effective permeability and breakthrough curves, the absence of a physical reason for these higher permeabilities warrants further consideration. If there are aperture distributions to match against flow as well, an analysis similar to that presented here could be used to determine which correlation model is most plausible for a given site. Explicitly, when trying to honor aperture (T) distributions, it may not be possible to also match observed flow properties, which would identify poor quality models.

The prevalence of flow channeling indicates that there is a large percentage of these sparse networks which is fairly stagnant, and that percentage is increased when a correlation between size and transmissivity is included in the model setup. The transport of solutes in these stagnant regions could result in increased matrix diffusion, retention and fracture-matrix exchanges due to the additional amount of time that solutes would remain there. Therefore it is not unreasonable to expect that reaction of a solute with the fracture walls would be highly variability throughout the network and that variability would depend on the assumed correlation between size and transmissivity. Through the use of particle tracking methods and the analysis tools used here it should be possible to identify these flow channeling pathways, as well as their complement, to potentially investigate their field relevance and perhaps identify them a priori for a given hydraulic setup. The approximation of active surface area was a first step in this direction and we are currently designing additional studies to investigate such properties.

Acknowledgments

Data: Model parameters for the generation of the DFN are adapted from Table 6-75 in the SKB report TR-10-52 [Svensk Kärnbränslehantering AB, 2010] and provided in Table 1 of this manuscript. The dfnWorks software suite can be obtained by emailing dfnworks@lanl.gov. Input files necessary to reproduce these experiments are available from the authors upon request (jhyman@lanl.gov). We would like to thank Jan-Olof Serloos for conversations and comments on the manuscript, Jean-Raynald de Druzy for his helpful review of the manuscript, and Daniel Fernández-García for handling the manuscript. JDH gratefully acknowledges the support of the U.S. Department of Energy through LANL Director's Postdoctoral Fellowship. JDH performed a portion of this work while visiting the Fraunhofer ITWM via funding provided by Intepore and he is grateful for their support as well. SK thanks NETL's SCNGO for their support. HSV, JDH, and SK acknowledge the support of LANL/LDRD project 20140002DR. NM thanks the DOE Used Fuel Disposition campaign for their support.

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